MOC (Management of Change) On-line Training



CONTENTS

1. Training Objectives.
2. What Is Management of Change (MOC)?
- Definition
- Core Purpose.
- Regulatory Basis
- Why MOC Matters:

3. What Constitutes a "Change"?
- Definition of Change
- The Strict RIK Exception
- Types of Changes Requiring MOC
- The Domino Effect of Change

4. Standard MOC Process (7 Steps)

5. Change Classification & Tiered Management.
    - By Time Dimension
    - By Risk Level
    - Elements of effective MOC Program
6. Roles and Responsibilities
7. Common Failure Modes and Pain Points
8. Key Learning Points
9. Practical Implementation & Recommendations
10. Discussion Questions & Exam

1. Training Objectives

By the end of this training, participants should be able to:

- Explain the definition, purpose, and regulatory basis of Management of Change (MOC)
- Identify which changes require an MOC using established criteria
- Apply the standard MOC process steps in practical scenarios
- Understand roles and responsibilities within the MOC process
- Analyze real-world case studies to recognize MOC failure patterns and their catastrophic consequences
- Recognize common MOC failure modes and implement improvement strategies

2. What Is Management of Change (MOC)?

2.1 Definition

Management of Change (MOC) is a formal, structured process used to assess and manage the risks introduced by changes to equipment, processes, procedures, or organizational structures — ensuring that safety, health, and environmental risks are controlled when a company makes changes in their facilities, documentation, personnel, or operations .

2.2 Core Purpose

The core goal of MOC is to ensure that all changes to a process are properly reviewed and that hazards introduced by the change are identified, analyzed, and controlled prior to resuming operation . Change represents a deviation from the original design, fabrication, installation, or operation of a process. Even simple changes, if not properly managed, can result in catastrophic consequences .

2.3 Regulatory Basis

In the United States, MOC is a mandatory requirement under OSHA's Process Safety Management (PSM) standard.

The standard requires employers to establish and implement written procedures to manage changes (except for "replacements in kind") to process chemicals, technology, equipment, and procedures, and changes to facilities that affect a covered process .

The MOC procedures must address the following considerations prior to any change :

The MOC procedures must address the following considerations prior to any change :

Required Consideration	Description
• Technical basis	• The rationale for the proposed change
• Safety and health impact	• How the change affects safety and health
• Operating procedure modifications	• Necessary updates to procedures
• Time period	• The necessary duration for the change
• Authorization requirements	• Who must approve the change

Additionally, employees involved in operating a process and maintenance and contract employees whose job tasks will be affected by a change must be informed of and trained in the change prior to start-up of the process or affected part

2.4 Why MOC Matters

Lack of management of change is one of the systemic problems that remain underlying causes of many incidents, even decades after major disasters .

Other recurring problems include lack of awareness of reactive hazards, inadequate plant design and maintenance, ineffective employee training, ineffective emergency preparedness, and lack of root cause incident investigations .

In more than 80% of major loss incidents, changes were involved. This section will demonstrate this through detailed case studies.

3. What Constitutes a "Change"?

3.1 Definition of Change

Change = any modification that is NOT a "Replacement-in-Kind" (RIK), including addition or removal of equipment.

3.2 The Strict RIK Exception

RIK means replacement that exactly meets original design specifications — no change in function, capacity, or operating conditions. This is a narrow exception. Any deviation from original design specs requires MOC.

3.3 Types of Changes Requiring MOC

Change Type	Examples	PSM Element Interconnections
• Equipment	• New machinery, piping modifications, valve replacement with different type	• Triggers updates to Process Safety Information, Process Hazard Analysis, maintenance procedures, training
• Materials	• Chemical substitution, catalyst replacement	• Requires hazard evaluation of new material properties
• Procedures	• SOP revisions, operating parameter adjustments	• Requires operator training before startup
• Process	• Capacity increases, process flow changes	• Triggers PHA revalidation
• Technology	• Control system software, System upgrades	• May require safety instrumented system verification
• Temporary	• Bypasses, temporary work areas	• Must have defined time limits and authorization
• Organizational/Personnel	• Staff reductions, reorganization, contractor changes	• Affects competency and continuity of process safety roles

- **Key Test:** Does the proposed change alter what is covered by Process Safety Information (PSI)? If yes, and it's not a true RIK, MOC applies.

MOC Triggers in Aviation Operations

Operational Area	MOC Trigger Examples
• **Flight Operations**	• New airport, new route, aircraft type introduction, procedure changes, automation upgrades
• **Ground Service**	• New ground handler, airport infrastructure changes, seasonal schedule changes
• **Cargo Operations**	• Aircraft substitution, new cargo types, loading procedure changes
• **MRO**	• New maintenance procedures, tooling changes, shift pattern changes, contractor introduction
• **Training Center**	• New aircraft type, syllabus revisions, instructor changes, simulator updates
• **Maintenance Control**	• Definition changes, reporting criteria updates, organizational restructuring
• **Management**	• Mergers, acquisitions, staff reductions, budget cuts, outsourcing decisions

Common MOC failures in Aviation

1. Communication Across Boundaries

Every case study reveals communication failures — between shifts, departments, management and operators, or parent company and subsidiary. MOC must explicitly address how changes are communicated across organizational boundaries.

2. Time Pressure and Procedural Shortcuts

Time pressure leads to shortcuts that bypass MOC. The MOC process must be designed to withstand schedule pressure.

3. Organizational Changes Are Safety Changes

Organizational changes — mergers, staff reductions, resource constraints — have direct safety consequences that must be managed through MOC.

4. Verification Is Essential

The load sheet officer assumed loading was complete; the maintenance supervisor assumed the replacement aircraft was safe; management assumed pilots understood MCAS. *Verification — not assumption — is the foundation of MOC.*

5. Normalized Deviance

When procedures are routinely bypassed without consequence, they lose their protective value.

3.4 The Domino Effect of Change

A practical example from OSHA training materials illustrates the interconnected nature of change :

During a routine inspection, a maintenance worker discovers a valve that no longer meets the applicable code and must be changed. Because the type of valve is no longer made, a different type of valve must be selected and installed. The type of valve selected may mandate different steps for the operators, who will require training and verification in the new procedures.

The rationale for selecting the type of valve must be made available for review by employees. When the new valve is installed, it will involve shutting down part of the process, so the employer must review the response plan to ensure procedures are adequate. After the new valve is in place, Process Safety Information must be updated before the Process Hazard Analysis is updated or revalidated. Inspection and maintenance procedures and training will also need to be updated.

This example demonstrates that one small change cascades through multiple PSM elements, which is why formal MOC procedures are essential

4. Standard MOC Process (7 Steps)

Step 1: Initiate Change Request
The requester submits a written request stating what is changing, why, and which PSI is affected.

Step 2: Hazard & Risk Assessment
Use analysis tools appropriate to the complexity — from checklists and What-If analysis to Hazard and Operability Study.

Step 3: Review & Approval
Subject matter experts review the request. Document who approved, when, and any conditions attached.

Step 4: Planning & Scheduling
Develop an implementation plan defining scope, responsible persons, timeline, and interim safety measures. Temporary changes must have authorized time limits.

Step 5: Implementation

Execute per the approved plan. Update PSI, P&IDs, and procedures simultaneously.

Step 6: Pre-Startup Safety Review (PSSR)

Before startup, confirm: construction matches design, procedures are updated, training is complete, and all MOC requirements are met.

Step 7: Closeout & Training

Train affected employees, complete documentation, and close the MOC.

5. Change Classification & Tiered Management

5.1 By Time Dimension

Type	Definition	Key Requirement
• **Permanent**	• Change implemented permanently	• Full MOC process
• **Temporary**	• Has a preset expiration (typically ≤90 days)	• Must be reverted or converted to permanent upon expiry; requires authorization and time limit
• **Emergency**	• Implemented rapidly due to urgent safety, environmental, or equipment integrity needs	• Expedited review with post-implementation documentation

9.2 By Risk Level

Not all changes need the same depth of review. The level of review should be commensurate with risk:

- **Level I:** Basic checklist (applies to all MOCs)
- **Level II:** Expanded checklist + What-If brainstorming
- **Level III:** HAZOP or equivalent methodology

6.3 Elements of an Effective MOC Program

According to Washington State VPP Best Practices, an effective MOC program includes :

- Procedures that consider the technical basis, safety/health impact, procedure modifications, time needed, and authorization required
- Steps to identify hazards before changes are made
- Methods to screen and classify changes
- Keys to identifying hazards in changes
- Methods for documenting MOC reviews
- Procedures to make approvals and authorizations workable
- Plans to inform and train personnel about changes
- Methods for updating Process Safety Information, procedures, and other PSM information

Roles & Responsibilities

Role	Core Responsibilities
• **Change Initiator**	• Identify change need, submit change request with technical basis
• **MOC Coordinator**	• Guardian of the MOC process, manages MOC system/software
• **Technical Reviewer**	• Conducts hazard identification and risk analysis
• **Approver**	• Makes approve/reject decisions based on risk assessment
• **Affected Employees**	• Participate in reviews, confirm understanding before operating changed equipment; must be trained prior to startup
• **Contract Employees**	• Must be informed of changes affecting their job tasks

7. Common Failure Modes & Pain Points

The most prominent MOC implementation issues include:

- **Failure to recognize** changes or initiate MOC **(the #1 pain point)**
- **MOC backlogs** — delayed or incomplete deliverable updates
- **Temporary changes** exceeding authorized time limits
- **Unclear roles** and responsibilities
- **Insufficient communication** — stakeholders not informed
- Overly **complex processes** — people don't understand why MOC matters
- **MOC closeout without completing updates** and follow-up activities

Other systemic problems that persist across industries :

- *Lack of awareness of reactive hazards*
- *Inadequate plant design and maintenance*
- *Ineffective employee training*
- *Ineffective emergency preparedness and community notification*
- *Lack of root cause incident investigations*

8. Key Learning Points

- **No change is "too small" to evaluate** — small changes can have cascading effects through interconnected PSM elements.
- **Temporary changes deserve the same rigor as permanent ones** — uncontrolled temporary changes often become permanent.
- **The RIK exception is extremely narrow** — any deviation from original design specs requires MOC
- **MOC requires multidisciplinary collaboration** — a single perspective cannot fully identify change-related hazards.
- **Leadership commitment is the foundation of MOC effectiveness** — BP Texas City demonstrated how a "culture of blindness to major risk" leads to disaster.

- **Organizational changes are changes** — personnel reductions (Bhopal), mergers (BP-AMOCO), and budget cuts can degrade safety performance.
- **Reactive hazards must be part of MOC assessment** — Bhopal demonstrated that failure to understand chemical reactivity can be catastrophic.
- **Emergency preparedness must be reassessed when changes occur** — Piper Alpha showed that emergency systems can fail when not maintained.
- **Process safety indicators must complement personal safety metrics** — BP's reliance on injury rates masked major process risks.
- **Lessons from past disasters must be actively taught** — "I am concerned the lessons are being forgotten and new engineers are unaware of the tragedy"

9. Practical Implementation Recommendations

- **Keep the process simple but applicable** — complex systems get bypassed
- **Build broad buy-in** — seek input from all affected groups during system development
- **Field-test before full implementation** — early debugging prevents long-term problems
- **Provide adequate training** — everyone involved in MOC should understand their role
- **Monitor effectiveness regularly** — integrate performance metrics
- **Use audits and management reviews** — unreviewed management systems eventually degrade
- **Ensure contractor coordination** — Toulouse (2001) demonstrated that subcontracting can lead to disengagement and loss of control

Discussion Questions:

- In your work area, what "changes" may be occurring that haven't been identified as requiring MOC?
- Are there temporary changes that have exceeded their original authorization period?
- What is the weakest link in your team's MOC process?
- If you discovered someone bypassing the MOC process to implement a change, what should you do?

Ethiopian Airlines MOC procedure and format: GSMSM Appendix-X.01

Ethiopian የኢትዮጵያ	ETHIOPIAN AIRLINES MANAGEMENT OF CHANGE FORM

MOC Ref. Number: _____

Date MOC Conducted: -_____

1. What is/are the source/s of the change/s?

Describe the source of change including but not limited to VISION 2035, Strategic business plan, New requirement, New facility, new regulatory body requirement, new license requirement, new training requirement, new station opening, new organizational structure change, new expansions, etc.

2.What is the change?

Describe the change including timescales

3. Where is the change [Owner of the change]?

- *Detail who is responsible to implement the change and who needs to be involved.*
- *Consider who it affects individuals, departments and organizations? Who needs to be notified of the change?*

Ethiopian Airlines MOC procedure and format:

4. Define the major components or activities of the change.

This will help you identify the main risks of each component or activity that will be populated in table 7 below.

5. What is the impact of the change?

Consider why the change is taking place and the impact on the organization and its processes and procedures. Will it impact the safety culture? Does it meet all regulatory requirements?

6. What follow up action is needed? (Assurance)

Consider how the change will be communicated and whether additional activities such as audits are needed during the change and after the change has taken place.

Ethiopian Airlines MOC procedure and format:

7. Safety Issues and the risk assessment

Hazard Risk ID No.	What is the issue? *Hazard*	What could happen as a result? *[Consequences]*	How Bad will it be? *[Severity]*	How likely is it to occur? *[Likelihood]*	Risk Rating [RI]	What action(s) are we taking? *[Mitigations]*	Action by whom and when [Owner]
	1					*There may be more than one action for each issue*	
	2						
	3						
	4						
	5						
	6						

8. Acceptance Confirmation

The identified risks are considered tolerable, and change is acceptable to implement			
Final Risk Mitigation Acceptance	Safety Auditor	Signature	Date
	Name:		
The management of change processes and procedures have been followed and the change can be implemented			
Implementation Acceptance	Mgr. QA and Safety	Signature	Date
	Name:		
	MD/VP/Airport CEO/EAU président	Signature	Date
	Name:		
	Manager Group SMS, IS & ERP	Signature	Date
	Name:		
	Director Group QMS, SMS, ERP & Compliance	Signature	Date
	Name :		

Ethiopian Airlines MOC procedure and format:

Appendix-A Change Prompts- Examples

Process	Any new and/or major business Policy, Process, Procedure, working instruction and SOP changes affecting the business model of the company. Ex. Strategic Business Plan
System and enhancements	New and/or changes in software systems, IT infrastructure, and technological tools supporting operations. Ex. Core Systems affecting the safety of flight like Q-Pulse, SMS systems, Maintenix, FDM analysis, SAP, flight planning systems, EFB, Schedule/rostering systems, Navigation systems, Baggage reconciliation systems, etc.
Resource Planning	New and/or changes in planning and allocation of human, financial, and physical resources to meet operational demands and compromising safety implementations and practices triggered by a change.
Facilities	Changes or new developments of physical infrastructure such as terminals, hangars, warehouses, buildings, maintenance shops, lounges, catering facilities, etc.
GSE/Equipment	Changes or acquisition of new GSE tools and equipment used to support airline operations. Ex. All GSE equipment, training devices (Mock-ups or simulators), cabin service carts or trolleys, ULDs, onboard safety equipment, cabin safety cards or videos, security screening equipment, onboard safety equipment (PBE, Life Jackets), etc.
Training	New or changes to training programs or curriculums for airline staff, crew, ground personnel, and technicians. Ex. New license and/or certification requirements.
Recruitment	New and/or changes in recruitment strategies, staffing levels, or talent acquisition methods.
Organization	New and/or major organizational structure changes or shifts in leadership, reporting lines, or decision-making processes. Ex. Expansion, Contraction, etc.
Regulation	New and/or changes in relation to compliance with local, national, and international regulatory requirements and standards.
Safety	Risk management strategies to ensure operational safety and prevent accidents. Ex. Vendor changes, etc.
Projects	New and/or major management of large-scale, strategic initiatives or capital project changes affecting safety of the flight.
Financial	New and/or major changes in financial planning, budgeting, and forecasting for airline operations affecting safety of the flight.
Commercial	Strategies related to revenue generation, marketing, and sales affecting safety of the flight. Ex. New routes, new charter flights, business booming/incremental operation changes, flight resumptions, etc.

Ethiopian Airlines MOC procedure and format:

Appendix-B Criteria for Safety Significance of the Change

Criteria	Brief Description
Potential Consequence of the change	If the most credible worst-case scenario for failure events occurs considering the current existing safety risk mitigations, MOC shall be conducted.
Novelty in the Change	MOC shall be conducted. • If the change is new only to the airline, • if the change has been implemented by industry elsewhere, • If the change is industry wide.
Complexity of the Change	MOC shall be conducted if the change results uncertainty in the potential consequences in case of failure i.e., The greater the complexity, the greater uncertainty.
Airline Monitoring Capabilities	If the airline monitoring capability to control the impact of the implemented change during operations is limited, this may trigger conducting MOC to be a must. That means, the lower the airline capability to control the impact of the change during operations, the higher the safety impact on operations. So, conducting MOC and addressing the new hazards if any is indispensable before resuming the operation.
Reversibility of before the change situation	It is the ability to restore a system or process to its original state before the change is implemented. If the new change leads to significant safety concerns or negatively impacts operation, and reversibility is not possible; conducting MOC is a must or vice versa.

Case Study 1: Malaysia Airlines A330 — Pitot Probe Covers Left On (Brisbane, 2018)
Sector: Flight Operations / Line Operations

Incident Overview

On 18 July 2018, a Malaysia Airlines Airbus A330 took off from Brisbane Airport with covers still installed on all three of its pitot probes (airspeed sensors). As a result, the flight crew received no airspeed indications from early in the take-off roll. The aircraft was operating a scheduled passenger service .

What Happened

The ATSB investigation revealed that the operator had recently reintroduced flights to Brisbane after a period of absence. When resuming operations at the airport, the operator's change management process had omitted many elements of established practices that were in place at Brisbane prior to the withdrawal .

The pitot probe covers (small tube-shaped sensor) — which are installed by ground engineers when an aircraft is parked to prevent insect nesting — were not removed before departure. Standard practice at Brisbane required specific verification steps, but the operator's MOC process failed to incorporate these established local practices into its procedures when it resumed operations at the port.

MOC Failures Analysis

MOC Element	Failure
• **Change identification**	• Resuming operations at a previously served airport was not treated as a significant change requiring full MOC assessment
• **Procedure adaptation**	• Existing Brisbane-specific practices were not incorporated into the operator's procedures
• **Risk assessment**	• Local operational hazards (pitot cover installation requirements) were not re-evaluated
• **Training/communication**	• Ground and flight crews were not adequately briefed on Brisbane-specific procedures
• **Verification**	• Pre-flight checks that would have detected the covers were not effectively implemented

Key Lessons

- **Resuming operations at an airport is a change requiring MOC. Even if the airline previously operated there, the absence period may have introduced new practices, personnel, or conditions that must be assessed.**

- **Local procedures must be integrated into MOC assessments. What works at one airport may not apply at another; MOC must capture site-specific requirements.**

- **Multiple layers of defense can be defeated by a single MOC failure. The pitot covers were not detected by pre-flight walk-around, flight deck checks, or ground crew — all because the procedural framework for Brisbane operations had gaps.**

Discussion Questions

- What specific elements should have been included in the MOC assessment for resuming Brisbane operations?
- How could the operator have verified that Brisbane-specific practices were correctly incorporated?
- What role should the ground handling agent play in the MOC process when an airline resumes operations at an airport?

Case Study 2: Ground Handling — Cargo Loading Communication Failure (Sydney, 1997)
Sector: Ground Service / Cargo Operations

Incident Overview

- On the morning of 16 October 1997, a Boeing 767-200 (ZK-NBC) was being prepared at Sydney Airport for a flight to New Zealand. When one engine was found unserviceable, passengers and their baggage were offloaded and placed on another flight. Two cargo containers were left in the hold — one secured and one unsecured — and the hold doors were left open .

- When a change occurs to an aircraft's departure time, a delay checklist is used by operations staff to ensure all departments are aware of the new time. However, due to deficiencies in the delay checklist, several departments of the ground handling company were not advised of the new departure time .

- The aircraft was eventually towed to the maintenance area for engine repairs. The towing crew likely closed the hold doors at this time. The decision to move the aircraft was made without advising Ramp Control, who were responsible for loading. Later, a new departure time of 2300 hours was planned, but Ramp Control was not advised — they assumed the aircraft would depart the following morning and did not allocate a loading team .

- The load sheet officer received information that led him to believe the additional cargo container would be loaded. The flight crew arrived to sign the load sheet. The load sheet officer assumed the containers had been loaded and signed the load sheet without verifying. The aircraft took off with an unsecured container in the hold, which moved during flight .

MOC Element	Failure
• **Communication of changes**	• Delay checklist failed to notify all relevant departments of the new departure time
• **Coordination**	• Aircraft moved without advising Ramp Control (responsible for loading)
• **Verification**	• Load sheet signed without verifying actual loading
• **Procedure adherence**	• Published loading procedures were bypassed due to time pressure (curfew)
• **Role clarity**	• Confusion about who was responsible for ensuring loading matched documentation

- **The ATSB identified significant factors including poor communications between departments, the load sheet officer's failure to verify loading, and priorities that placed departure before curfew ahead of adherence to published loading procedures**

Safety Actions Taken

- The ground handling agent implemented several corrective actions :
- Reviewed procedures for inter-departmental communications
- Reviewed the delay checklist
- Introduced new procedures requiring the loading supervisor to sign the load instruction form upon completion
- Issued written instructions to all ramp and load control staff
- Installed additional monitors in load control and ramp control offices

Key Lessons

- Delay checklists are critical MOC tools. When departure times change, the communication system must ensure all affected departments are informed.
- Assumptions are dangerous. The load sheet officer assumed loading was completed; verification is essential.
- Time pressure can erode procedural compliance. The curfew deadline contributed to shortcuts that bypassed safety procedures.
- Inter-departmental communication is a known weak point. MOC must explicitly address how changes are communicated across organizational boundaries.

Discussion Questions

- What specific changes to the delay checklist would have prevented this incident?
- How should the MOC process handle "last-minute" changes that occur under time pressure?
- What cultural factors contributed to the load sheet officer's decision to sign without verifying?

Case Study 3: Maintenance — Shift Turnover Failure Leads to Fatal Crash (1991)
Sector: MRO / Aircraft Maintenance

Incident Overview

A commuter airline aircraft underwent maintenance work the night before a fatal accident. During the second shift, the scheduled removal and replacement of both left and right horizontal stabilizer leading edge/deicer boot assemblies was undertaken. Each leading edge required removal of more than 40 screws from each assembly. Two mechanics and an inspector were assigned .

One crew member removed screws from the top side only on the left-side leading edge. A shift change took place before the work was completed. The bottom screws were never removed from the left-side leading edge, and — critically — the upper screws were not replaced. There was no entry in the work turnover form of the status of work performed, nor was the new supervisor verbally informed of the partial removal .

- The right-side leading edge was subsequently replaced by the next shift, and both upper and lower screws were reinstalled. The supervisor decided to postpone replacing the left-side leading edge assembly due to limited time. The aircraft was moved out of the hangar and signed back into service .

- The flight crew for the morning departure was not informed about the work performed. The top of the horizontal stabilizer cannot be seen during a normal pre-flight walk-around inspection. The airplane was assigned to a passenger flight. During the return leg, the loss of the leading edge's aerodynamic surface caused an almost immediate stall of the horizontal stabilizer. There were no survivors among the 14 persons aboard.

MOC Failure Analysis

MOC Element	Failure
• **Shift turnover**	• No written record of work-in-progress; no verbal briefing to incoming supervisor
• **Procedure adherence**	• Manual required written records and briefings — these were not followed
• **Work status communication**	• Flight crew not informed about incomplete maintenance
• **Release to service**	• Aircraft signed back into service with known incomplete work
• **Supervision**	• Confusion among supervisors about shift turnover responsibilities

The investigation found that some practices in the airline's hangar were not being performed in accordance with the General Maintenance Manual (GMM) and that these practices reflected "accepted procedures" — meaning the deviation had become normalized.

Organizational Context

- The investigation also revealed that the airline had undergone a merger of two large commuter airlines and assets of a third (Britt Airways, Rocky Mountain Airways, and Bar Harbor Airlines) to form a new entity. The FAA Principal Maintenance Inspector indicated that these merger events considerably limited his time for on-site inspections, and the depth and quality of inspections were limited by lack of time.

Key Lessons

- Shift turnover is a critical MOC moment. When work crosses shift boundaries, formal communication is essential.
- Work-in-progress must be documented. The manual required written records; shortcuts defeated this safeguard.
- Flight crews must be informed of maintenance work. The pre-flight walk-around cannot detect all maintenance issues (e.g., T-tail top surface).
- Organizational changes (mergers) can degrade oversight. The merger reduced inspection resources, contributing to the failure to detect procedural deviations.
- Normalized deviance is dangerous. When procedures are routinely bypassed, they lose their protective value.

Discussion Questions

- What specific shift turnover procedures would have prevented this accident?
- How should MOC account for organizational changes like mergers that affect safety oversight?
- What signs indicate that a "normalized deviation" is developing in a maintenance organization?

Case Study 4: Cargo Operations — Last-Minute Aircraft Swap (UPS Flight 2976, 2025)
Sector: Cargo Operations / Flight Operations

Incident Overview
On 4 November 2025, UPS Flight 2976, a McDonnell Douglas MD-11 freighter, crashed shortly after takeoff from Louisville Muhammad Ali International Airport, killing three crew members and 12 people on the ground. Another 23 were injured. The flight was bound for Hawaii .

What Happened
The aircraft that crashed (N259UP) was not originally intended to make the trip. A different MD-11 (N260UP) was the original assigned aircraft. During pre-flight inspection, a crew member discovered a fuel leak and reported it to maintenance, which deemed the aircraft not airworthy .

The last-minute aircraft swap placed the crew on a different plane. Cargo was moved from the old plane to the new plane "without incident," and the crew spent approximately two hours reviewing a new briefing package and operational flight plan

- As the same crew member who discovered the fuel leak visually inspected the new MD-11, he again encountered the same load supervisor and joked about "meeting again." The face-to-face maintenance brief was described as "uneventful, with good natured banter" .

- The NTSB investigation revealed that the aircraft's left engine and pylon assembly separated from the wing immediately after takeoff rotation. Surveillance footage showed the engine detaching before the plane entered an unrecoverable descent. Investigators found cracks in a critical bearing within the engine mount assembly that were not noticed during routine maintenance checks .

- Records showed the aircraft carried a long operational history and had undergone prior maintenance tied to its engine mounting structure. Investigators identified at least ten prior cases involving related cracking issues on similar aircraft. Boeing had previously advised operators to replace problematic spherical bearings but did not require mandatory corrective action

MOC Failure Analysis

MOC Element	Failure
• **Aircraft substitution**	• Last-minute swap introduced a different aircraft with different maintenance history
• **Maintenance history review**	• The replacement aircraft's engine mount history was not adequately assessed before assignment
• **Risk assessment**	• The change in aircraft was not treated as a significant MOC event requiring full evaluation
• **Communication**	• The light-hearted nature of the maintenance brief suggests the seriousness of the swap was underestimated
• **Inspection adequacy**	• Critical components had not undergone detailed inspection since 2021

- **The NTSB hearing focused heavily on whether warning signs about the MD-11's engine mounting system were overlooked for years. Similar flaws involving the same component had been documented multiple times over the past 15 years, yet only a fraction were formally reported to the FAA. Regulators stopped short of issuing an airworthiness directive before the crash .**

Key Lesson

- **Aircraft swaps are major MOC** events. Substituting one aircraft for another introduces a new set of maintenance histories, operational limitations, and risk profiles.
- **Maintenance history must be reviewed** before assignment. The replacement aircraft's known issues with engine mounts should have triggered enhanced inspection or rejection.
- **"Good natured banter" can mask risk**. The informal tone of the brief may have contributed to inadequate scrutiny of the replacement aircraft.
- **Known fleet-wide issues require fleet-wide action**. Ten prior cracking cases should have triggered mandatory inspection requirements.
- **Aging aircraft require enhanced oversight**. The MD-11 fleet's engine mount issues were known but not adequately addressed.

Discussion Questions

- What specific MOC procedures should be triggered when an aircraft is substituted at the last minute?
- How should known fleet-wide maintenance issues be incorporated into aircraft assignment decisions?
- What cultural factors might explain why a maintenance brief became "good natured banter" rather than a serious risk review?

Case Study 5: Flight Training — B747 Training Program Delays (Ansett, 1990s)
Sector: Training Center / Flight Operations

Incident Overview

Ansett Airlines introduced Boeing 747 aircraft to its fleet, requiring a comprehensive training program for flight crews. The training program's start was delayed by a month, which had cascading effects on the entire project. The flight operations management modified the flow chart used to plan the training program when the delay occurred.

What Happened

The original plan and its modification did not incorporate details of tasks distinct from, but related to, the priority task of training crews. These tasks included :

- Development of relevant manuals for flight crew
- Hiring and indoctrination of contract crews
- Securing air operators certificates and other approvals from the Civil Aviation Authority.

- Because these tasks were not included in the original plan, any consequences of the delays were not highlighted and could not be monitored against planned milestones .

- The late start of the training program resulted in the need to extend the role of contract crews to include training and checking. However, no adjustment to the flow chart was made to account for added time needed for training or checking the skills of contract crew members in training roles .

- The task of organizing recruitment and indoctrination of contract crews was allocated outside the development group to managers in the domestic operations area. This group did not have knowledge of the B747 and had insufficient time to develop a training program that fully addressed the contract crew members' indoctrination needs .

- The flight operations development team leader did not fully utilize available resources for planning or implementation. Several experienced non-management pilots were knowledgeable in both international and B747 operations, but their knowledge and experience were not used during the planning process .

MOC Failure Analysis

MOC Element	Failure
• **Change planning**	• Delay to training start did not trigger reassessment of all dependent tasks
• **Resource allocation**	• Managers without B747 knowledge were assigned critical training development tasks
• **Expertise utilization**	• Experienced pilots were not consulted during planning
• **Project control**	• Development team leader lost effective control when tasks were delegated outside the team
• **Milestone monitoring**	• Related tasks not in the original plan could not be tracked

- **The investigation found that "planning was flawed, the indoctrination process for contract crew members did not adequately prepare them for their allocated tasks, and manuals were not developed in a structured manner"**

Regulatory Context

- **The Civil Aviation Authority also contributed to the situation. Inadequate resources in the Melbourne office led to deficient handling of the approval process for the air operators' certificates. The need to use type-rated specialist inspectors from Sydney, the loss of the project manager, and poor communications between regional offices added to confusion regarding approvals for manuals and training syllabuses .**

Key Lessons

- **Changes in training timelines cascade**. A delay in one phase (training start) affects all dependent tasks (manual development, crew recruitment, regulatory approvals).

- **Planning must include all dependent tasks**. The original flow chart's failure to include manual development, crew hiring, and regulatory approvals meant these tasks were invisible to milestone tracking.

- Expertise must be utilized. **Experienced pilots were available but not consulted** — a critical resource waste.

- **Project management requires centralized control**. When tasks were delegated outside the development team, coordination broke down.

- **Regulatory approval is a critical** path item. Failure to account for CAA approval timelines contributed to the compressed training schedule.

Discussion Questions

- What specific MOC procedures should have been triggered by the one-month delay?
- How should a training organization plan for regulatory approval timelines in an MOC process?
- What are the risks of assigning critical training development tasks to managers without subject matter expertise?

Case studies Summary

Case Study	Sector	Primary MOC Failure	Key Lesson
• **Malaysia Airlines A330 (Brisbane)**	• Flight Operations	• Airport resumption without full MOC	• Resuming operations is a change requiring local procedure integration
• **B767 Cargo Loading (Sydney)**	• Ground Service / Cargo	• Delay checklist deficiencies; communication breakdown	• Change communication across departments is critical
• **Continental Express Maintenance**	• MRO	• Shift turnover failure; no work-in-progress record	• Work crossing shifts requires formal handover
• **UPS Flight 2976**	• Cargo / Flight Ops	• Last-minute aircraft swap without adequate assessment	• Aircraft substitutions are major MOC events
• **Ansett B747 Training**	• Training Center	• Delay cascading; expertise not utilized	• Training timeline changes require full dependent task reassessment

Thank You!